

to connect to other devices in the kitchen and generate useful solutions. The most fundamental step of which would be for the fridge to monitor and maintain its own systems so users are not left in the lurch when it suddenly stops working. The self-diagnostic features measure the fridge's performance with regards to temperature and energy consumption, and senses if it's not performing at full potential. This irregularity is then passed on to the owner with the one-touch option of registering a maintenance call that fits into the user's schedule. This is useful even on a warning level which can help ensure that timely action is taken and no food goes to waste due to spoilage. In addition, other features like sending a personal alert to the user on their smartphone or smartwatch letting them know if the door has been left open for longer than usual or if the fridge needs to be defrosted would be essential.

But more than these simple communications, the fridge of the next generation can do so much more. By equipping fridges with barcode sensors and scanners, we can let the fridge keep track of the expiry dates of all its contents. This would result in users being warned when items are about to expire and even present an option for creating a grocery list which can be used when the user is out shopping. Imagine sending a message to your fridge asking it what needs to be brought? It isn't that unreal a scenario given the technology that exists. The fridge can even be trained to recognise preferred brands, food items, flavours and eating patterns, and be authorised to conduct online shopping for essential items autonomously without ever bothering the user.

The interactivity of the device with others in your home is also useful. For example, being able to transfer a recipe from a TV show to the fridge's memory would quickly run a cross check to see which ingredients are missing and execute a command for the next shopping run. Alternatively, another feature is the ability of the fridge to make recommendations of what to cook based on the items currently in stock - not only would this save time in searching and figuring out what to cook but also allow for new experiences that compliment your tastes (as learnt by the fridge). The potential for the interactivity takes us into a whole other world of implementations which we shall discuss next.

Intelligent cooking and cleaning

The fridge is but one device. Once it is connected with the cooking appliances such as a microwave, electric gas, oven and other gadgets things can